

Kathiravan B

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CAREER OBJECTIVE

AI and Data Science graduate with hands-on experience building production-grade backend microservices, containerized ML deployments, and cloud applications. Highly proficient in Java, Python, Spring Boot, distributed systems, and AI integration. Seeking a Software Development Engineer (SDE), Backend Engineer, or AI Engineer role to build scalable infrastructure and deliver immediate impact on core product teams.

EXPERIENCE

Freelance Research Intern

Jul 2026 – Aug 2026

Freelance

Chennai, Tamil Nadu

- Designed and delivered a full-stack gym management platform for a luxury fitness brand, engineering both internal control panels and customer-facing interfaces for real-time attendance tracking, client/trainer management, and operational dashboards.
- Architected a high-throughput event notification engine using FastAPI, Next.js, and MongoDB Atlas, integrating Brevo Email API and external cron workers to automate transactional emails, billing alerts, and scheduling workflows.
- Implemented robust security protocols by building custom JWT-based authentication and granular role-based access control (RBAC) to enforce strict data isolation across client, trainer, and administrative portals.

SDE

Dec 2025 – Mar 2026

RocketLane

Chennai, Tamil Nadu

- Built a multi-language transcription pipeline using WhisperX, improving word-level alignment accuracy by 25%; packaged the model as a Docker container and deployed it on AWS SageMaker with inference endpoints wired into backend services.
- Constructed backend infrastructure for meeting video playback and transcript UI, enforcing standardized video storage, signed URL access, transcript normalization across 3 providers, and closed-caption synchronization.
- Improved system reliability by stabilizing Zoom Meeting SDK recording pipelines and fixing duplicate bot lifecycle issues; set up configurable Datadog monitoring with custom thresholds, reducing undetected failures by 50%.
- Shipped production features including GDPR recording consent, calendar rescheduling fixes, and an MCP server proof-of-concept for Chargebee integration; tracked down and fixed several backend bugs that were causing data mismatches across services.

AI Research Intern

Jun 2025 – Jul 2025

BluBridge

Chennai, Tamil Nadu

- Studied and executed Large Language Model internals including architecture, tokenization, attention mechanisms, and fine-tuning strategies through hands-on research tasks, reducing model debugging time by 30%.
- Engineered Distributed Data Parallel (DDP) and multi-GPU training pipelines, cutting training time by approximately 40% compared to single-GPU runs and improving compute utilization across the team.
- Applied transformer interpretability techniques such as embedding visualization and attention heatmaps to trace and debug model decision patterns, identifying 3 key failure modes in the custom LLM.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript

Frameworks & Technologies: Flask, FastAPI, React, Node.js, Spring Boot, LangChain, RESTful APIs

Databases: MongoDB, PostgreSQL, MySQL, NoSQL, Vector Databases (Qdrant, FAISS)

DevOps & Cloud: Docker, Containerization, AWS SageMaker, Google Cloud, GitHub, Version Control, Datadog

Areas: Backend Development, Software Engineering, Data Processing, Microservices, System Design, MCP Servers, Information Technology, Scalable Systems, Deployment, Statistics, Probability

Soft Skills: Problem-Solving, Communication, Teamwork, Leadership, Adaptability, Critical Thinking, Collaboration, Training

Certifications: Machine Learning, Data Science, Deep Learning, Full Stack Development

PROJECTS

AI-Powered YouTube Filtering Extension | *Python, NLP, JavaScript, Chrome Extensions API*

- Built a Chrome extension that filters and recommends YouTube videos based on user-defined prompts and time-based sessions.
- Leveraged NLP and machine learning to semantically analyze video metadata including title, description, captions, and comments.
- Engineered real-time API handling, user session tracking, and dynamic UI injection into the YouTube DOM.

Home Fitness Tracker & Recommendation Web App | *Node.js, Express.js, React, MongoDB*

- Designed and deployed a personalized home fitness platform with user authentication, workout recommendations, performance tracking, and daily activity logs.
- Integrated third-party APIs for exercise datasets and instructional videos, generating dynamic, adaptive fitness plans.

Cold Mailer: RAG-Driven Automated Email Generator | *Python, FastAPI, Streamlit, LangChain, RAG, SMTP*

- Developed a local RAG-based cold email generation platform using an asynchronous FastAPI backend and a Streamlit frontend interface.
- Engineered a localized knowledge base pipeline utilizing LangChain to parse and ingest PDF materials, enabling context-aware, hyper-personalized email generation.
- Integrated an automated SMTP dispatch layer alongside a dynamic template persistence and prompt history management system to optimize outreach tracking.

Recruitment Analytics Platform | *Python, Flask, MongoDB, Qdrant, Docker*

- Designed and built a data-driven recruitment system to automate candidate evaluation, cutting manual shortlisting effort by 50%.
- Constructed pipelines for PDF parsing, text embedding, and Qdrant-based vector similarity search.
- Developed REST APIs for candidate scoring against job requirements; deployed as a containerized solution for scalability.

EDUCATION

St. Joseph’s Institute of Technology <i>Bachelor of Artificial Intelligence and Data Science – CGPA: 8.51</i>	Chennai, Tamil Nadu 2022-2026
Montfort Matriculation Higher Secondary School <i>HSC: 89% SSLC: 87%</i>	Chennai, Tamil Nadu

Relevant Coursework: Software Development, Machine Learning, Deep Learning, Data Science, Python, Core Java

CODING SKILLS

LeetCode: Solved 200+ problems spanning arrays, dynamic programming, graphs, and trees; consistently practicing to strengthen algorithmic thinking and problem-solving speed.

GeeksForGeeks: Solved 100+ problems covering data structures, sorting algorithms, and competitive programming concepts to reinforce core CS fundamentals.

HackerRank: Solved 100+ problems across domains including Python, SQL, and problem-solving tracks, earning 6400 Hackos.

SkillRack: Ranked 2300 with 1000 Bronze medals and 2000+ problems solved.